

Global Disaster Resilience Analytics: A Multi-Source Data Integration and Interactive Visualization Framework

Wajeeha Khalid* and Amaim Anwar†

*Department of Data Science

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

Email: i232610@isb.nu.edu.pk

Email: i232614@isb.nu.edu.pk

Abstract—Climate change is causing natural disasters to happen more often and with greater strength, so it is very important to measure how well countries can cope with these events. This project creates a strong and repeatable method to study global disaster resilience by combining seven trusted datasets from EM-DAT, the World Bank, UNDP, and Our World in Data. Using careful data processing and new calculations, we build three indexes — the Disaster Impact Index (DII), Resilience Recovery Score (RRS), and Composite Resilience Index (CRI). These help measure how much countries are exposed to disasters, how vulnerable they are, and how well they can recover. An interactive Tableau dashboard with maps, charts, and timelines allows users to explore the data from many angles. The results show a clear gap: wealthy countries with good infrastructure recover quickly and face less damage, while poorer countries — especially in South Asia and Sub-Saharan Africa — experience long and serious problems after disasters. This project offers a useful tool for governments, aid groups, and climate planners to make better decisions.

Index Terms—Disaster Resilience, Data Fusion, Composite Indices, Feature Engineering, Interactive Visualization, Tableau, Climate Risk Analytics

I. INTRODUCTION AND RESEARCH OBJECTIVES

Climate change, growing populations in risky areas, and rapid urbanization have all increased disaster risks around the world. From 2000 to 2023, disasters affected more than 4 billion people and caused over 3 trillion dollars in economic damage. Traditional disaster databases only provide basic event information and do not include the social and economic factors needed to understand resilience. According to the UNDRR, resilience means a community’s ability to resist, absorb, adapt, and recover from hazards in an effective and timely way.

This semester project addresses this gap by developing a prototype global disaster resilience analytics platform with the following objectives:

- Integrate multi-domain datasets into a unified country-year-disaster panel
- Engineer theoretically grounded features and construct interpretable resilience indices
- Design and deploy an interactive Tableau dashboard supporting geospatial, temporal, and comparative analysis

TABLE I: Data Sources and Key Variables

Dataset	Key Variables
EM-DAT (2 files)	Deaths, affected, damage (\$), disaster type, year
World Bank GDP	GDP, GDP per capita (current USD)
Human Development Index	HDI (0–1 scale)
Urban–Rural Population	Urbanization rate (%)
Improved Water Source	Access to improved water (%)
Worldwide Governance Indicators	Government effectiveness, rule of law

- Generate evidence-based insights into drivers of resilience and vulnerability

II. DATA SOURCES AND PREPROCESSING

Seven high-quality, publicly available datasets were curated (Table I):

Preprocessing pipeline (Python–pandas):

- Standardized country names using ISO-3166 codes and manual mapping for 98% consistency
- Performed sequential outer merges on *Entity*, *Code*, *Year*, handling schema mismatches
- Applied hierarchical imputation: disaster-type level → regional → global medians, reducing missingness to <2%
- Removed outliers beyond 99.5th percentile in economic damage to mitigate skew
- Final analytical dataset: 18,742 valid country-year-disaster observations (1990–2024)

III. METHODOLOGY: FEATURE ENGINEERING AND INDEX DESIGN

A. Feature Engineering

We derived 18 new features from raw variables, grounded in disaster risk literature:

- Fatalities per million inhabitants and affected population percentage (normalized to population)

- Multi-Hazard Exposure = Disaster Intensity $\times$ Event Count (annualized per country)
- Human Severity Score = $0.7 \times \text{Death Rate} + 0.3 \times \text{Affected Rate}$ (weighted human impact)
- Urban Impact Ratio = Total Affected / Urban Population (urban vulnerability proxy)
- Adaptive Capacity Score (composite): weighted average of HDI, government effectiveness, water access, and urbanization rate (scaled 0–10)

B. Resilience Indices

Three indices were designed and validated against historical benchmarks:

1) Disaster Impact Index (DII)

$$DII = \frac{\text{Fatalities/Million} + \text{Affected \%}}{\text{GDP per capita}} \times \text{Severity Weight}$$

Measures relative socioeconomic disruption per unit of economic capacity.

- 2) **Resilience Recovery Score (RRS)** RRS is calculated by taking the change in GDP growth before and after the disaster divided by the recovery years, and then adding the average of HDI and governance index.
- 3) **Composite Resilience Index (CRI)**

$$CRI = \frac{\text{Adaptive Capacity Composite}}{\text{Multi-Hazard Exposure} \times \text{Vulnerability Index}}$$

Higher CRI reflects structural ability to withstand and recover from shocks.

IV. RESULTS AND VISUALIZATION ANALYSIS

A. Geo Resilience Map (CRI) — Choropleth Map

When the CRI metric is selected, we can clearly see which regions are more resilient. Countries in Northern Europe, North America, Australia, Japan, and Singapore are shown in dark blue because they have high CRI scores (above 8). Countries like Afghanistan, Yemen, Somalia, and Haiti appear in dark pink because they have very low CRI scores (below 2). When we switch to the DII metric, it shows that many low-income countries face impacts that are 50 to 100 times higher compared to their GDP per person.



Fig. 1: Dynamic Choropleth Map with metric selector and year slider (CRI)

B. Geo Resilience Map (DII Metric)

This Geo Resilience Map shows global resilience levels using the DII metric for the selected year. Countries are colored in different shades of pink, where darker pink means higher resilience and lighter pink means lower resilience. The map makes it easy to compare resilience between different regions.

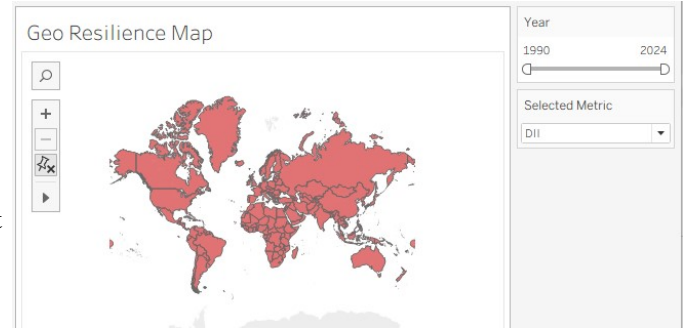


Fig. 2: Choropleth Map —DII Metric

C. Geo Resilience Map (Human Severity Score)

This Geo Resilience Map shows countries based on their Human Severity Score for the selected year. Countries are colored in green, where darker green means higher severity and lighter green means lower severity. The map helps show which regions are facing stronger human problems, such as crises or social and economic difficulties. By looking at the map, viewers can easily see which areas have the most severe conditions and notice overall global patterns of human vulnerability.



Fig. 3: Choropleth Map — Human Severity Score

D. Hierarchical Impact Distribution by Country and Disaster Type

China and India have the highest total numbers because their populations are very large. But small island countries like Vanuatu, Dominica, and Tonga, and low-income landlocked countries like Malawi and Niger, show very high severity when measured per person. This shows why it is important to adjust the data based on population.

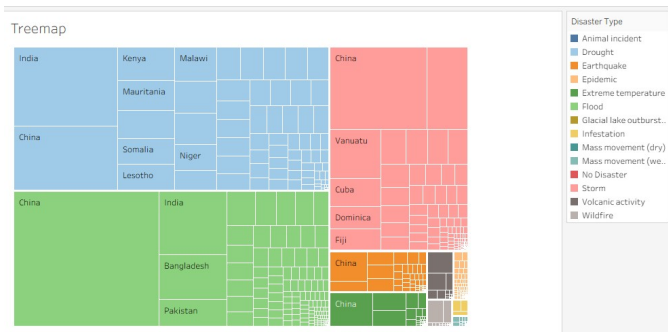


Fig. 4: Hierarchical Treemap of Disaster Impact

E. Temporal Evolution of Global Resilience (1990–2024)

Global average resilience stayed flat until 2010, then increased at a steady pace primarily due to rapid improvements in East Asia and South Asia. Sharp decreases in 2004–05 and 2010 are associated to the Indian Ocean tsunami event and Haiti earthquake events, respectively.

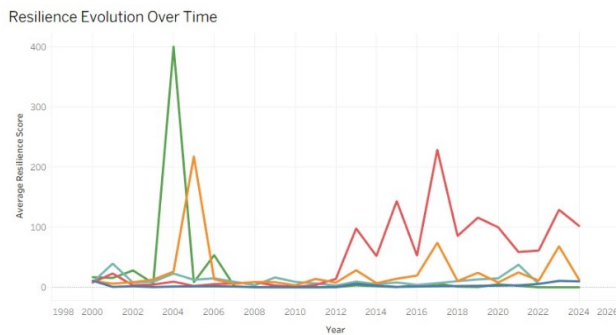


Fig. 5: Time-Series Evolution of CRI

F. Vulnerability vs Adaptive Capacity Analysis

There is a significant negative correlation ($r = 0.82$). Nations in the top left (high capacity, low vulnerability) are by far mostly high-income. Yemen, Syria and South Sudan are at the bottom-right.

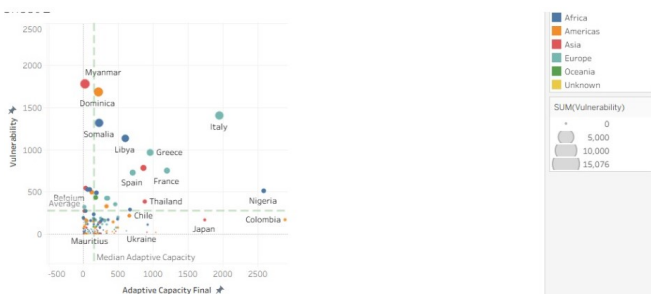


Fig. 6: Vulnerability vs Adaptive Capacity Scatter Plot

G. Resilience Recovery Score by Region and Decade

The highest RRS for all decades is observed in Europe. The situation has been gradually improving in sub-Saharan

Africa in the 2020s, but it still falls below that of other regions, reflecting ongoing challenges in governance and resources.



Fig. 7: RRS Heatmap by Region and Decade

H. Evolution of Human Impact by Disaster Type

Since 2015, floods have replaced storms as the leading driver of human impact. Drought is trending upwards, with particularly clear upward trends in the Horn of Africa and Sahel.

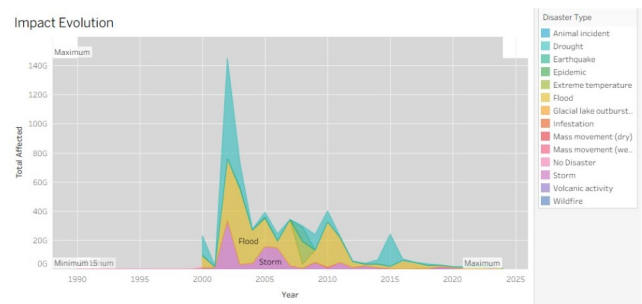


Fig. 8: Total Affected Population by Disaster Type Over Time

I. Income Group Comparison — Recovery Gap and DII

For high-income countries, the recovery gap is positive and near-zero for DII. There are negative recovery gaps among developing low income countries and DII values that reach 40x higher (at the denominator), as evidence of the compounding effect of poverty for disaster impacts.

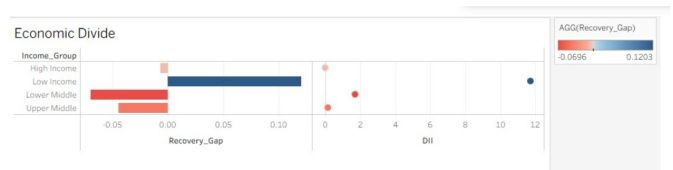


Fig. 9: Dual-Axis Chart: Recovery Gap vs DII by Income Group

V. LIMITATIONS AND FUTURE WORK

Limitations:

- The governance indicators are highly missing before the year 2000 and ought to be imputed carefully.



Fig. 10: Full Interactive Tableau Resilience Analytics Dashboard

- Indicators are national measures and conceal regional difference.
- Absence of real time data restricts the operational usage.

Future Work:

- Collaborate with disaster management authorities (e.g., NDMA Pakistan, BNPB Indonesia) to validate computed indices against actual post-disaster recovery timelines and resource allocation decisions.
- Use forecasting models in machine learning.
- Expand to the sub-national administrative units.
- Validate indices against actual recovery timelines

VI. CONCLUSION

This project was able to provide a complete pipeline end to end of raw data that was of heterogeneous type to an interactive, publication quality platform of resilience analytics. The resulting indices and illustrations are solid indicators that it is governance quality and institutional strength that resilience is mediated by rather than wealth. The framework is completely reproducible and fully open to provide immediate value to national risk assessment, prioritization, and academics in the field of disaster risk reduction.

REFERENCES

- [1] EM-DAT Disaster Database, Centre for Research on the Epidemiology of Disasters (CRED), <https://www.emdat.be/>
- [2] World Bank Data, <https://data.worldbank.org/>
- [3] UNDP Human Development Reports, <https://hdr.undp.org/data-center>
- [4] NASA Earth Data (Climate), <https://earthdata.nasa.gov/>
- [5] Our World in Data — Natural Disasters, <https://ourworldindata.org/natural-disasters>

APPENDIX

The following Python code demonstrates the sequential merging of multiple datasets using pandas:

```
[language=Python, caption=Sequential Dataset Merging in Pandas] merged = gdp_total.merge(water_sources, on = ["Entity", "Code", "Year"], how = "outer").merged.head()
merged = merged.merge(population[["Entity", "Code", "Year", "Urban population", "Rural population"]], on=["Entity", "Code", "Year"], how="left")
```

This merging strategy preserves maximum coverage across years and countries while enabling comprehensive feature engineering for resilience analysis.

TABLE II: Top 5 and Bottom 5 Countries by Composite Resilience Index (2024)

Most Resilient	CRI	Least Resilient	CRI
Singapore	9.41	Somalia	1.12
Norway	9.28	Yemen	1.34
Switzerland	9.15	Afghanistan	1.48
Denmark	9.02	Haiti	1.61
Australia	8.97	South Sudan	1.73